MIL-C-81524(AS)
31 March 1967
Superseding
WS 4378
11 June 1965

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 82 MOD 0

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This document governs the manufacture, assembly and preparation for delivery of the MARK 82 MOD 0 impulse cartridge and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-129	Marking for Shipment and Storage

FSC 1377

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DRAWINGS

Bureau of Naval Weapons

DL 2518360	Cartridge, Impulse, MARK 82 MOD 0

PUBLICATIONS

BUSANDA Instruction 4030.17	Preservation, Packaging, Packing and Marking; Navy Policy for

Bureau of Naval Weapons

OP 5	Ammunition Ashore (This publication is not furnished to bidders and contractors by the procuring activity. Contractors should contact the Government Inspector for use of the publication.)
WR-43A	Preparation of Quality Assurance Provisions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78	Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite the latest edition and supplements thereto.)

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3. REQUIREMENTS

3.1 General.

3.1.1 The MARK 82 MOD 0 impulse cartridge covered by this specification shall be manufactured in accordance with DL 2518360 and all documents listed thereon.

3.2 Material. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referred to on the respective Bureau of Naval Weapons drawings unless specific approval in writing covering a departure therefrom has been obtained from the Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of 70 cartridges shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.3 Primary components. For the purposes of this specification the main propellant charge and the first increment of propellant are defined as primary components. See 4.3.3.

3.4 Ceramic-to-metal seal. The ceramic-to-metal seal shall be in accordance with Bureau of Naval Weapons Drawing 2518363 of DL 2518360. See 4.7.2.

3.5 Insulation resistance. Prior to assembly of the bridgewires, the insulation resistance shall not be less than 50 megohms at a relative humidity of less than 80 percent when measured as specified in 4.7.1.

3.6 Bridgewire. The bridgewire shall be soft soldered to the pins.

3.7 Bridgewire assembly, continuity. The assembly shall be thoroughly cleaned to remove flux and contamination. Subsequent to cleaning, the bridgewires shall be flashed by application of an electric current to assure continuity of both bridges. The current shall be regulated to produce a visible red glow of the bridges.

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An EMF of approximately 1.5 volts AC or DC may be used with a suitable series resistor to adapt the source characteristics to the bridge resistance. Flashing is a necessary manufacturing operation and shall be conducted on each bridgewire assembly. Assemblies which fail to glow in both of the bridges shall be screened from acceptable material in the process. The faulty assemblies may be reworked by replacement of the bridges.

3.8 Igniter composition and loading. The igniter composition shall be compounded and loaded in accordance with Bureau of Naval Weapons Drawings 1558439 and 2518365 of DL 2518360.

3.8.1 X-ray. All ignition elements shall pass X-ray examination of 4.7.3.

3.8.2 Black powder. The moisture content of the black powder at the time of loading shall be less than 0.90 percent as determined by the test procedure of 4.7.4.1.

3.9 Propellant charges.

3.9.1 Quantity. All propellant charges shall be as specified on Bureau of Naval Weapons Drawing 2518402 of DL 2518360. Charges shall be determined by weight rather than by volume measurements.

3.9.2 Smokeless powder, handling and loading. Do not subject the smokeless powder to unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.9.2.1 Smokeless powder, condition. The smokeless powder utilized shall exhibit no evidence of decomposition. See 4.7.4.3.

3.10 Change of propellant lot. Whenever a change occurs in the propellant lot, validity of the charge determination shall be verified by performing the test prescribed in 4.10.

3.11 Cartridge inspection.

3.11.1 Visual inspection. The complete cartridge shall be free of the visible defects noted in 4.7.6.

3.11.2 Bridgewire resistance. The resistance of each bridge in each completely assembled cartridge, when measured electrically as specified in 4.7.5 shall be not less than 0.08 ohm nor more than 0.14 ohm.

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3.11.3 X-ray. All cartridges shall pass X-ray examination of 4.7.7.

3.12 Ballistic requirements. Cartridges from each production lot, selected in accordance with 4.7.8.1, shall meet the requirements listed below when subjected to the test prescribed in 4.7.8.

3.12.1 Maximum pressure. The upper and lower limits of maximum pressure shall be U = 8500 psi and L = 5800 psi, respectively.

3.12.2 Misfire. There shall be no misfires. See 4.7.8.2.

3.12.3 Hangfire. There shall be no hangfires. See 4.7.8.3.

3.13 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes and coatings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-414. Unless otherwise specified in the contract, order, or requisition, all samples for test purposes shall be provided at the expense of the contractor.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with MIL-STD-414.

4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 70 cartridges shall be prepared. Sixty of these cartridges shall be expended in the tests listed below and 10 shall be retained for investigative purposes. A

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preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of 3.2 through 3.11.3 of this specification. The contractor shall deliver the cartridges to the activity designated by the contracting agency for test. Preproduction sample acceptance shall consist of the following tests as prescribed in Table 3 of MIL-D-21625: (Number in parentheses denotes number of cartridges to be allocated for each test) -65°F (12), +70°F (12), +160°F (12), vibration (12), and 15 "g" shock (12). No cartridge shall produce results outside the limits set forth in 3.12.1 through 3.12.3. Firings shall be conducted in the test fixture of 4.7.8.1. Any quantity production prior to notification by the contracting agency of pre-production sample acceptability shall be at the contractor's risk. The preproduction cartridges will not be applied as a part of the quantity specified for delivery by the contract.

4.3.3 Production lot. The suggested production lot size is 181 minimum and 1300 maximum cartridges. Cartridges required for all test purposes are to be selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract. Only primary components from a single lot shall be used in a production lot of cartridges. One primary component production lot may be used in more than one cartridge production lot. Cartridges selected for the ballistic test of 4.7.8 shall be delivered to the activity designated by the contracting agency.

4.4 Gages. The contractor shall provide himself with gages as necessary to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43A. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics, by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and esaminations which are not annotated with a classification code are classified minor.

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4.6 Production and acceptance inspection. Production and acceptance inspection shall consist of verification of the characteristics classified on the drawings of DL 2518360, and the following tests and examinations.

Item	Nature of Test	Requirement	Test	Classification (WR-43A)
Primary Component	Single lot verification	3.3	4.3.3	(M101)
Ceramic-to-Metal Seal	Bond verification	3.4	4.7.2	(M102)
Pins and Body Assembly	Insulation resistance	3.5	4.7.1	(M103)
Body and Bridgewire Assembly	Soldering verification	3.6	---	(M104)
Body and Bridgewire Assembly	Continuity	3.7	---	(C1)
Ignition Charge	Igniter composition and loading	3.8	---	(C2)
Ignition Element	X-ray	3.8.1	4.7.3	(C3)
Black Powder	Moisture content	3.8.2	4.7.4.1	(C4)
Propellant Charge	Type and quantity verification	3.9.1	---	(C5)
Smokeless Powder	Condition	3.9.2.1	4.7.4.3	(C6)
Cartridge	Ballistics (change in propellant lot)	3.10	4.10	(C7)
Cartridge	Visual inspection	3.11.1	4.7.6	(M105)
Cartridge	Bridgewire resistance	3.11.2	4.7.5	(C8)
Cartridge	X-ray	3.11.3	4.7.7	(C9)
Cartridge	Ballistics	3.12	4.7.8	(C10)

4.6.1 Packing, packaging and marking. (M106) The inspector shall ascertain that the packing and packaging of the cartridges and the container markings conform to Section 5 of this specification.

4.7 Tests.

4.7.1 Insulation resistance. Prior to assembly of the bridgewires, the insulation resistance shall be checked with an applied voltage of 500 ± 25 volts DC. The measurement shall be made from shorted pins to the ignition element body. Units which fail to meet the requirements of 3.5 shall be rejected.

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4.7.2 Ceramic-to-metal seal. The ceramic-to-metal seal for each cartridge shall be tested in accordance with Bureau of Naval Weapons Drawing 2518363 of DL 2518360 prior to bridgewiring.

4.7.3 X-ray of ignition element. Each ignition element, prior to insertion in the cartridge, shall be mounted in suitable racks or frames and oriented so that the most revealing exposure can be obtained. Pressure sensitive tape may be used as an aid in positioning the ignition element. Reject all ignition elements that have any questionable qualities such as: missing or defective spacer or abnormal or missing lead styphnate ignition charge.

4.7.4 Black powder and smokeless powder.

4.7.4.1 Moisture content of black powder. Conduct at least one moisture test for each cartridge lot or daily (whichever requires the more frequent testing) and such additional tests as considered warranted by variations in conditions or process. Samples may be collected at the site of loading from the black powder container, or taken from cartridges prior to sealing the black powder charge in the cartridges. Weight losses per booster charge shall be ascertained within an accuracy of 1 percent. The presence of moisture in excess of the limitations of 3.8.2 shall cause suspension of loading until corrective measures have been taken.

4.7.4.2 Determining moisture content of black powder. Enclose a powder sample in a tared container. Weigh and expose for drying at 90°C ± 5°C for 1-1/2 hours. Close the container, cool and weigh. The loss of weight is calculated as the percentage of moisture in the original sample.

4.7.4.3 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable portions of paragraphs 1308-1318 and 1322 of OP 5. Evidence of decomposition shall be cause for rejection.

4.7.5 Bridgewire resistance. Conformance with the bridgewire resistance of 3.11.2 shall be ascertained by means of a test circuit which limits the bridge current to 25 milliamperes, maximum. The test circuit shall be connected to the cartridge through a fully assembled Bendix connector, PC 06E-8-4S or equal. A new connector shall be used after a maximum of 500 connections. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable bridge which is known to be accurate within 1 percent in the range of resistance specified shall be used to determine whether or

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not the requirement is being met. Units which fail to meet the requirements of 3.11.2 or fail to assemble completely, by hand, with the specified connector shall be rejected. The results of bridgewire resistance measurements shall not be affected by resistance of the test circuit and connector.

4.7.6 Visual inspection. Examine each cartridge produced under a contract and reject for dents, deep scratches, split or cracked lip in the crimped area, perforated closure discs, sharp edges, burrs, incomplete threads and any other defects which may conceivably prevent entry into the firing chamber or affect stability or resistance to moisture.

4.7.7 X-ray examination. Examine all cartridges by X-ray and any observable imperfections in assembly shall be cause for rejection. X-ray plates may be destroyed after ballistic acceptance of cartridges.

4.7.8 Ballistic tests.

4.7.8.1 The number of cartridges selected from each production lot for the ballistic tests shall be in accordance with the requirements of Table A-2 of MIL-STD-414, Inspection Level IV and an AQL of 0.04. The cartridges shall be conditioned for 24 hours at +70°F ± 5°F and fired in the Aero 7A bomb ejector rack with a 2000 pound store attached and with an expended MARK 2 MOD 1 impulse cartridge in one chamber. The firing current shall be 5 amperes per bridge. A firing head, suitable for firing the MARK 82 MOD 0 impulse cartridge in the Aero 7A ejector rack is shown in Figure 1. A record of pressure versus time and of elapsed time from close of firing key to start of pressure rise shall be taken by suitable instrumentation for each firing. In addition to the sample of cartridges required for the above test, 10 additional cartridges shall be furnished from the production lot for investigative purposes.

4.7.8.1.1 Criteria of acceptability. The lot shall be acceptable if the requirements of 3.12.1 are met at an AQL of 0.04 under the provisions of Section B, Part II of MIL-STD-414.

4.7.8.2 Misfire. The failure of any cartridge to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.7.8.3 Hangfire. One or more hangfires shall result in rejection of the lot represented. A hangfire is defined as a firing in which the elapsed time between close of firing key and start of pressure rise exceeds 0.05 second.

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4.8 Retest. There shall be no retests.

4.9 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, original test results should be discarded and the complete test reconducted.

4.10 Change in propellant lot. Whenever a change in propellant lot occurs, validity of the charge determination shall be verified by firing 12 cartridges each at temperatures of -65°F, +70°F and +160°F in the test fixture of 4.7.8.1. Results of these firings shall fall within the limits prescribed in 4.3.2. If the cartridges fired at +70°F meet the requirements of 3.12 they may be counted as part of the sample required by 4.7.8.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. For definitions of packaging levels, see 6.4.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Four cartridges shall be packaged in an inner container in accordance with Bureau of Naval Weapons Drawing 2518378 of DL 2518360.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be packaged in accordance with Method III of MIL-P-116, and with the Code of Federal Regulations 49 CFR 71-78.

5.2 Packing. For definitions of packing levels, see 6.4.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Six inner containers as described in 5.1.1 shall be packed in accordance with Bureau of Naval Weapons Drawing 2518379 of DL 2518360.

5.2.2 Level B. Not applicable.

5.2.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests, and are packaged in

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accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

5.3 Markings.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

5.3.2 Normal markings. In addition to the markings required by contract or order, and as specified on Bureau of Naval Weapons Drawings 2518378 and 2518379 of DL 2518360, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The MARK 82 MOD 0 impulse cartridge was developed to serve as a replacement for the MARK 1 MOD 3 impulse cartridge where used in the escape hatch opener of the A3 type aircraft.

6.2 Ordering information. Procurement documents should specify the following:

a. title, number and date of this specification;

b. whether preproduction inspection is required and if so, specify the test activity;

c. production lot size and test activity (see 4.3.3);

d. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of Contractors and Contracting Officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents, need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NAVWEPS Form 4200/15 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

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6.3.1 Test report. Results of ballistic tests shall be forwarded to the Naval Air Systems Command and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. The definitions of Levels A and C packaging and packing given below were extracted from BUSANDA Instruction 4030.17.

6.4.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

6.4.1.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity, for immediate use. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.4.2 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.2.1 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations, and may be the supplier's commercial practice when such meets the requirements of this level.

6.5 Explosives safety precautions.

6.5.1 Applicability. The loading, assembly and handling of the explosives, subassemblies and the finished items covered by this specification, involve hazardous operations and therefore require suitable explosives safety precautions. At plants of the Naval Air Systems Command, Department of the Navy, and other Government plants where the Government is responsible for safety, the precautions listed below are mandatory. These same minimum provisions listed below are recommended for observance by a contractor or manufacturer that explosively loads the item covered in this specification in order to fulfill his own safety responsibility. Other warnings and precautions,

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pertaining to the operational effectiveness or safety of the specified item during its use (for example, during its handling, launching or firing) are included in the detailed technical requirements of the specification. Use of this specification shall not be construed to replace other safety regulations of governmental agencies or to relieve the contractor or manufacturer of responsibility for the safety of his operations.

6.5.2 Personnel and equipment. Proper care must be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In connection with the above, the precautionary measures in the following paragraphs should be observed. (See also paragraph 6.5.4, Static and electrical and other applicable parts of these safety precautions.)

6.5.2.1 Explosives operations shall be kept under strict supervision. Periodic inspections shall be employed to assure compliance with established safety regulations.

6.5.2.2 Safe equipment and methods should be utilized for handling and transporting explosives and loaded parts.

6.5.2.3 Where required, remote control barricaded handling equipment shall be used for explosives operations such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.5.2.4 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.5.2.5 Maintain equipment such as presses, dies, crimpers, blenders, mixers, sifters, conveyors, and the like, in safe operating condition.

6.5.2.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

6.5.2.7 During static tests, propulsion units such as rocket motors shall be restrained against full thrust to prevent flight if ignited, for example, during test of igniter circuit. Also, means to vent flames and exhaust gases harmlessly shall be provided.

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6.5.3 Operating procedures. All loading operations shall be conducted in full accordance with a formalized operating procedure which shall include the following:

6.5.3.1 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.5.3.2 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.5.3.3 Appropriate safety instructions for specific operations shall be posted and personnel instructed as to the hazards present and precautions involved.

6.5.3.4 The number of working personnel and transients permitted in each work space shall be established and posted in such areas. The numbers shall be the minimum number required to accomplish necessary operations.

6.5.3.5 All loading operations should be conducted in a neat and orderly manner.

6.5.3.6 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings.

6.5.3.7 Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from, floors, walls, ceiling, ledges, tables, benches, piping, and equipment or the items loaded. Clean up any spilled material immediately.

6.5.3.8 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.5.3.9 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which affect functioning should, insofar as practicable, avoid using bare fingers or improper equipment, in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

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6.5.3.10 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges. (See paragraph 6.5.4.)

6.5.4 Static and electrical.

6.5.4.1 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges shall be employed. Fabrics such as silk and nylon, which promote static generation shall be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low energy type electric primers, detonators and squibs. The latter types of items shall have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding from induced electric currents originated by sources such as lightning, static, electromagnetic radiations, etc. Where necessary for safety, humidity of work rooms should be appropriately increased, as required to lessen electrostatic effects but without inducing excessive moisture absorption by any of the components of the item being processed.

6.5.4.2 Protect all explosive operations from effects of electric current originating in equipment such as soldering irons or guns, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding, separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator or squib. For example, circuits may be inadvertently completed from a defective electric soldering iron through a grounded contact. All electric type primers, detonators or squibs provided with wire leads shall have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a

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finished item. Removable short circuiting clips, or other devices should be employed during manufacturing operations involving items such as electric primers, detonators or squibs.

6.5.4.3 Where necessary, protect personnel and operations from electromagnetic energy. In certain work with very sensitive explosives or items, a shielded environment free from significant electromagnetic energy should be provided.

6.5.5 Operating facility and storage regulations.

6.5.5.1 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance, at a minimum, with the American Table of Distances (ATD); and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with Intraplant Distances set forth in the ATD when outside of loading rooms. For Navy managed or contractor managed Navy owned explosives loading plants, the Armed Services Explosives Safety Board standards approved by the Department of Defense, the Secretary of the Navy, and the Naval Air Systems Command, covering quantity-distance regulations for explosives, apply.

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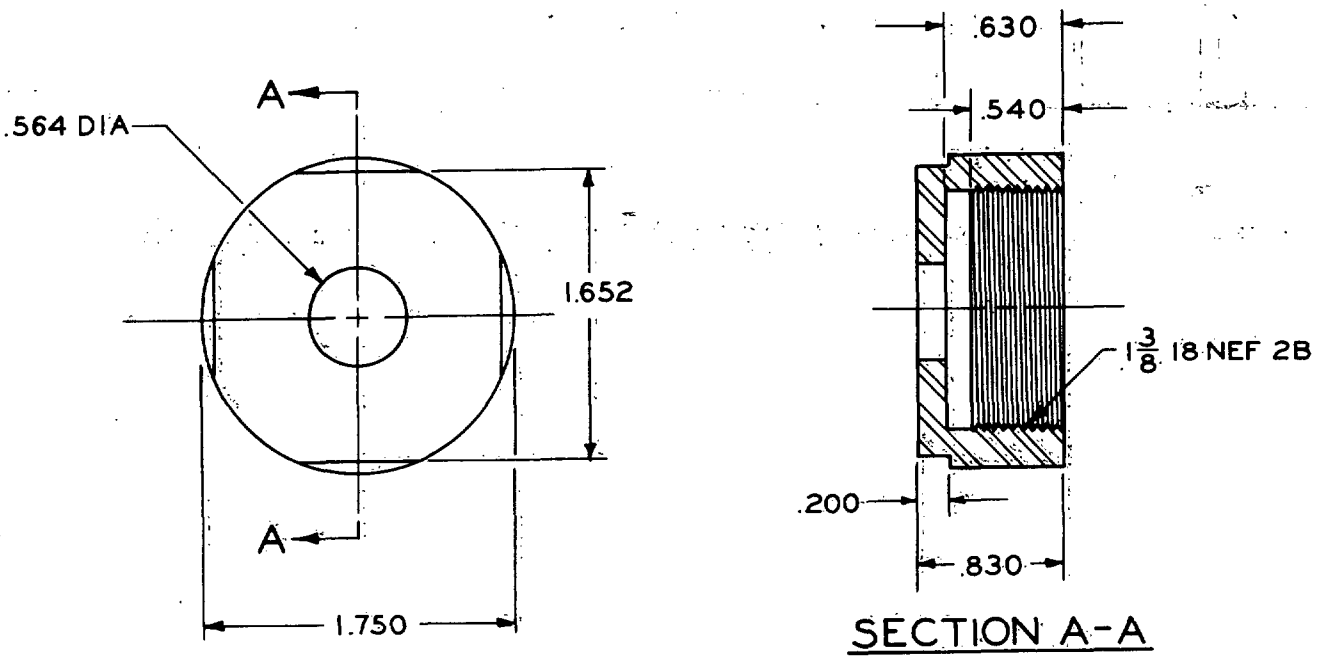


FIGURE 1 FIRING HEAD FOR FIRING THE MARK 82 MOD 0 CARTRIDGE IN THE AERO 7A BOMB EJECTOR RACK

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-81524(AS) CARTRIDGE, IMPULSE, MARK 82 MOD 0

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Naval Air Systems Command
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